


# APPLIED DATA STRUCTURES & ALGORITHMS LAB (BCS-553)

## B.Tech CSE 5th SEMESTER

**SUBMMITED TO:**

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# Lab Assignment 1: Reading Array Values

## Experiment No. 1.1 - Read first and last array values

### Experiment Title:

Read first and last array values

### Aim:

To read the first and the last value of an array using indexes.

### Software / Tools Required:

Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

### Theory:

An array stores values of the same type in consecutive indexed positions. Indexing starts at 0, so the first value is a[0] and the last is a[a.length - 1]. A for loop visits every index in turn, which lets a program read, print or add all the values.

### Algorithm / Procedure:

1. Create the array marks = {45, 67, 80, 52, 90}.
2. Print marks[0] with the label First.
3. Print marks[marks.length - 1] with the label Last.
4. Compile and run the program.

### Program / Queries / Commands:

```
// Name: Mankirat Singh
// ID: BCS2024542
// Roll No: 24103032571
// Assignment 1, Task 1
// Question: Create the array int[] marks = {45, 67, 80, 52, 90}. Print
// "First:" followed by the first value and "Last:"
followed by the last // value, on two lines, using indexes.

public class Main {     public
static void main(String[] args) {
int[] marks = {45, 67, 80, 52, 90};

        System.out.println("First: " + marks[0]);
        System.out.println("Last: " + marks[marks.length - 1]);
    }
}
```

### Input:

Fixed array declared in the program: marks = {45, 67, 80, 52, 90}.

## Output:

```
First: 45
Last: 90
```

**Time / Space Complexity:** Time O(1); Space O(1).

## Result:

The program was compiled and executed successfully. The first and last values of the array were read correctly using indexes.

## Viva-Voce Questions:

- What is an array index?
- What is the first valid index in Java?
- How do you find the last index of an array?

# Experiment No. 1.2 - Print all array values

## Experiment Title:

Print all array values

## Aim:

To print every value of an array on its own line using a for loop.

## Software / Tools Required:

Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

## Theory:

An array stores values of the same type in consecutive indexed positions. Indexing starts at 0, so the first value is a[0] and the last is a[a.length - 1]. A for loop visits every index in turn, which lets a program read, print or add all the values.

## Algorithm / Procedure:

1. Create the array marks = {45, 67, 80, 52, 90}.
2. Start a loop with i = 0 and continue while i < marks.length.
3. In each iteration print marks[i].
4. Compile and run the program.

## Program / Queries / Commands:

```
// Name: Mankirat Singh
// ID: BCS2024542
// Roll No: 24103032571
// Assignment 1, Task 2
// Question: Create the array int[] marks = {45, 67,
80, 52, 90}. Print // every value on its own line,
using a for loop.

public class Main {     public
static void main(String[]
args) {         int[] marks =
{45, 67, 80, 52, 90};

        for (int i = 0; i < marks.length; i++) {
System.out.println(marks[i]);
        }
    }
}
```

## Input:

Fixed array declared in the program: marks = {45, 67, 80, 52, 90}.

## Output:

```
45
67
80
```

```
52
90
```

**Time / Space Complexity:** Time O(n); Space O(1).

## Result:

The program was compiled and executed successfully. All values of the array were printed in order using a for loop.

## Viva-Voce Questions:

- What does marks.length return?
- Why does the loop start at 0?
- What is a loop iteration?

# Experiment No. 1.3 - Calculate array sum

## Experiment Title:

Calculate array sum.

## Aim:

To add all the values of an array using a loop and print the total.

## Software / Tools Required:

Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

## Theory:

An array stores values of the same type in consecutive indexed positions. Indexing starts at 0, so the first value is a[0] and the last is a[a.length - 1]. A for loop visits every index in turn, which lets a program read, print or add all the values.

## Algorithm / Procedure:

1. Create the array and set sum = 0.
2. Visit every element with a for loop.
3. Add each value to sum.
4. Print sum with the label Sum.

## Program / Queries / Commands:

```
// Name: Mankirat Singh
// ID: BCS2024542
// Roll No: 24103032571
// Assignment 1, Task 3
// Question: Create the array int[] marks = {45, 67,
80, 52, 90}. Add all // the values with a loop and
print the total as "Sum:" followed by the // total.

public class Main {     public
static void main(String[]
args) {         int[] marks =
{45, 67, 80, 52, 90};
int sum = 0;

        for (int i = 0; i <
marks.length; i++) {
sum = sum + marks[i];         }

        System.out.println("Sum: " + sum);
    }
}
```

## Input:

Fixed array declared in the program: marks = {45, 67, 80, 52, 90}.

## Output:

```
Sum: 334
```

**Time / Space Complexity:** Time O(n); Space O(1).

## Result:

The program was compiled and executed successfully. The first and last values of the array were read correctly using indexes.

## Viva-Voce Questions:

- What is an accumulator variable?
- Why is sum initialised to 0?
- What is the sum of the given array?